Linear Simultaneous Equations using Substitution

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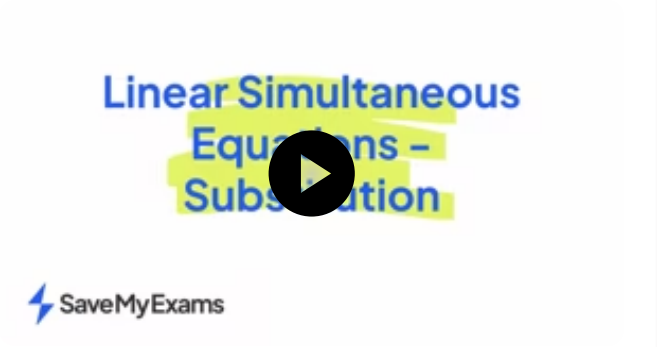

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Help me understand

Linear simultaneous equations using substitution

What are linear simultaneous equations?

- When you have more than one equation in more than one unknown, then you are dealing with **simultaneous equations**
- An equation is **linear** if none of the unknowns in it is raised to a power other than one
- Solving a **pair of simultaneous equations** means finding **pairs of values** that make **both equations true** at the same time
- A linear equation in two unknowns will produce a straight line if you graph it... linear = line
- A pair of simultaneous equations will produce lines that will cross each other (if there is a solution!)

How do I use substitution to solve linear simultaneous equations?

Step 1: Rearrange one of the equations to make one of the unknowns the subject (if one of the equations is already in this

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

equation not used in Step 1

Step 3: Solve the new equation from Step 2 to find the value of one of the unknowns

Step 4: Substitute the value from Step 3 into the rearranged equation from Step 1 to find the value of the other unknown

Step 5: Check your solution by substituting the values for the two unknowns into the original equation you **didn't** rearrange in Step 1

Examiner Tips and Tricks

- Although elimination will always work to solve simultaneous linear equations, sometimes substitution can be easier and quicker.
- Knowing both methods can help you a lot in the exam (plus you will need substitution to solve quadratic simultaneous equations).

Worked Example

Solve the simultaneous equations:

$$3x + 5y = -9$$
$$y = 1 - 9x$$

LABEL THE EQUATIONS

$$3x + 5y = -9 \quad (1)$$
$$y = 1 - 9x \quad (2)$$

y IS ALREADY THE SUBJECT OF EQUATION (2), SO WE CAN SKIP STEP 1

STEP 2: $3x + 5(1 - 9x) = -9$
$3x + 5 - 45x = -9$
$-42x = -14$

SUBSTITUTE EQUATION (2) INTO EQUATION (1)

STEP 3: $x = -14 \div (-42) = \frac{1}{3}$

STEP 4: $y = 1 - 9(\frac{1}{3}) = 1 - 3 = -2$

USING EQUATION (2)

STEP 5: $3(\frac{1}{3}) + 5(-2) = -9$
$1 - 10 = -9$
$-9 = -9$ ✓

USING EQUATION (1)

$x = \frac{1}{3} \quad y = -2$

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